Lecture 32

Motion along a path

Uses.

Summary 8: Scalar differentiation, product, chain and inverse derivative rules hold under their stated hypotheses.
Summary 9: A positive derivative makes a real function strictly increasing on an interval.
Summary 12: Coordinates define the Euclidean dot product and norm; triangle and reverse triangle inequalities hold.
Summary 13: The cross product gives oriented planar area after embedding in $\mathbb{R}^3$, and $\|\boldsymbol{u} \times \boldsymbol{v}\| = \|\boldsymbol{u}\|\,\|\boldsymbol{v}\| \sin\theta$ for nonzero vectors.
Summary 14: Polar coordinates give $x = r\cos\varphi$, $y = r\sin\varphi$. The running ellipse has $r = p/(1 + \varepsilon\cos\varphi)$, $p > 0$ and $0 \le \varepsilon < 1$.
Summary 23: Two scalar primitives on an interval differ by a constant, fixed by one prescribed value.
Summary 26: The fundamental theorem differentiates accumulation integrals of continuous functions.
Summary 28: A C^1 parametrised curve has length $\int \|\boldsymbol{r}'\|$; polar area for one sweep is $\frac{1}{2}\int r^2\,\mathrm{d}\varphi$.

Objectives. *A path specifies position as a function of time. Its first two derivatives give velocity and acceleration, while prescribed acceleration and initial data reconstruct the motion. We then resolve those derivatives in a rotating polar basis and along the tangent and normal to the path.*

By the end of this lecture you will be able to:

- Recover a C^2 motion from continuous acceleration and initial position and velocity.
- Derive polar velocity, acceleration and signed area rate for a positive C^2 radius and a C^2 angle.
- Derive the tangent–normal acceleration formula for a regular C^2 path where curvature is positive.
- Compute curvature, and retain the separate zero-speed, zero-curvature and polar-origin guards.

32.1 Position and componentwise differentiation

Let I be a nonempty open real interval and let $d = 2$ or $d = 3$. A position function is a map $\boldsymbol{r} : I \to \mathbb{R}^d$. A vector limit means that the norm of the error tends to zero. For every vector

$\boldsymbol{z} \in \mathbb{R}^d$, each component satisfies $|z_j| \le \|\boldsymbol{z}\| \le \sum_{k=1}^{d} |z_k|$. Thus a vector limit exists exactly when all its component limits exist, with the same component values. Apply this observation to the difference quotient:

$$\boldsymbol{r}'(t) = \lim_{h \to 0} \frac{\boldsymbol{r}(t+h) - \boldsymbol{r}(t)}{h} = (r_1'(t), \dots, r_d'(t)).$$

The derivative is defined where this limit exists. A vector function is C^1 when its component derivatives exist and are continuous, and C^2 when the first and second component derivatives are continuous. A dot denotes a derivative with respect to time t.

For a C^2 motion, the *velocity*, *speed* and *acceleration* are

$$\boldsymbol{v} = \dot{\boldsymbol{r}}, \qquad v = \|\boldsymbol{v}\|, \qquad \boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{\boldsymbol{r}}.$$

Velocity is a vector; speed is its nonnegative norm. Differentiating finite component sums gives the scalar-product rule $(\boldsymbol{u} \cdot \boldsymbol{w})' = \boldsymbol{u}' \cdot \boldsymbol{w} + \boldsymbol{u} \cdot \boldsymbol{w}'$. Indeed, each term $u_j w_j$ has that product derivative, and summing gives the identity. Scalar-times-vector products obey the same rule componentwise.

Where $\boldsymbol{v}(t_0) \neq \boldsymbol{0}$, the *tangent line* is $\boldsymbol{r}(t_0) + \lambda \boldsymbol{v}(t_0)$, with $\lambda \in \mathbb{R}$. The derivative definition justifies its direction: $\boldsymbol{r}(t_0 + h) = \boldsymbol{r}(t_0) + h\boldsymbol{v}(t_0) + \boldsymbol{e}(h)$ with $\|\boldsymbol{e}(h)\| / |h| \to 0$. The line describes the first-order displacement; a zero derivative would not determine its direction by this formula.

32.2 Reconstructing motion from acceleration

A vector integral is defined by integrating each component. Let a continuous acceleration $\boldsymbol{a} : I \to \mathbb{R}^d$ and initial data $t_0 \in I$, $\boldsymbol{r}_0, \boldsymbol{v}_0 \in \mathbb{R}^d$ be given. Define velocity and position successively by

$$\boldsymbol{v}(t) = \boldsymbol{v}_0 + \int_{t_0}^{t} \boldsymbol{a}(s)\, \mathrm{d}s, \qquad \boldsymbol{r}(t) = \boldsymbol{r}_0 + \int_{t_0}^{t} \boldsymbol{v}(s)\, \mathrm{d}s. \tag{32.1}$$

Each finite interval between t_0 and t lies inside I. The scalar fundamental theorem in every component gives $\boldsymbol{v}' = \boldsymbol{a}$, $\boldsymbol{r}' = \boldsymbol{v}$ and the prescribed initial values. The resulting position is C^2. It is unique: two such velocities have zero derivative in each component, so their difference is constant by Lecture 23 and zero at t_0. The same argument applied to the positions gives zero difference everywhere. This establishes existence and uniqueness for the stated prescribed-acceleration problem.

For the next model, let $g > 0$ be the downward acceleration magnitude and $h > 0$ the initial height. Let (u, w) be the initial velocity, with $u, w > 0$. Figure 32.1 shows a representative case.

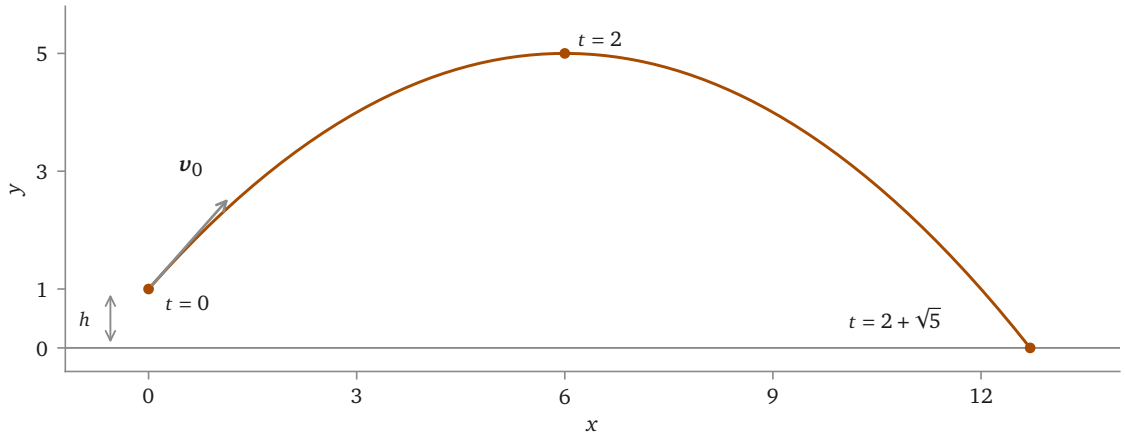


Figure 32.1: The model with $u = 3$, $w = 4$, $h = 1$ and $g = 2$ in chosen units. The initial-velocity arrow shows its direction. The marked times come from the exact trajectory below.

Example 32.1 (A projectile from its acceleration)**.** Assume a particle has constant acceleration $(0, -g)$, initial position $(0, h)$ and initial velocity (u, w) at $t = 0$, with $g, h, u, w > 0$. Resistance is neglected in this prescribed model. Find its position and the first positive time it reaches the line $y = 0$. Integrating the components and retaining both initial constants gives

$$\boldsymbol{v}(t) = (u, w - gt), \qquad \boldsymbol{r}(t) = (ut, h + wt - gt^2/2).$$

Direct differentiation and evaluation at zero verify the acceleration and both initial data, so uniqueness identifies the motion. The ground equation has roots

$$t = \frac{w \pm \sqrt{w^2 + 2gh}}{g}.$$

Because $\sqrt{w^2 + 2gh} > w$, exactly one is positive: $t_* = (w + \sqrt{w^2 + 2gh})/g$. The physical flight is $0 \le t \le t_*$. The vertical velocity vanishes at $t = w/g$, giving maximum height $h + w^2/(2g)$. The horizontal velocity remains $u > 0$. For the displayed numerical model, $t_* = 2 + \sqrt{5}$ and the highest point is $(6, 5)$ at $t = 2$.

32.3 The moving polar basis

Let $r : I \to (0, \infty)$ and $\varphi : I \to \mathbb{R}$ be C^2 functions giving the planar representation $\boldsymbol{r}(t) = r(t)(\cos\varphi(t), \sin\varphi(t))$. Use a continuous real angle, allowing complete turns instead of reducing it discontinuously to a half-open angle range. Define the *radial* and *azimuthal unit vectors* by

$$\boldsymbol{e}_r = (\cos\varphi, \sin\varphi), \qquad \boldsymbol{e}_\varphi = (-\sin\varphi, \cos\varphi).$$

Their norms are 1 and their dot product is zero by $\cos^2\varphi + \sin^2\varphi = 1$. Their orientation is counterclockwise from the first to the second.

Figure 32.2 shows this basis at one point. Its direction changes with the angle even when the scalar components remain fixed.

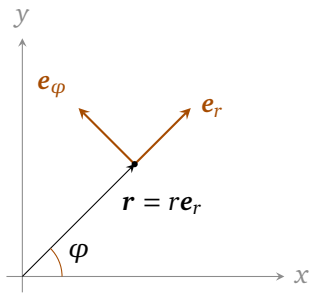


Figure 32.2: The radial vector points away from the origin. The azimuthal vector is its counterclockwise quarter-turn; both are attached to the moving point.

Componentwise chain differentiation gives $\dot{\boldsymbol{e}}_r = \dot\varphi \boldsymbol{e}_\varphi$ and $\dot{\boldsymbol{e}}_\varphi = -\dot\varphi \boldsymbol{e}_r$. For example, the first derivative is $(-\sin\varphi\dot\varphi, \cos\varphi\dot\varphi)$, exactly the stated multiple. Apply the product rule to $\boldsymbol{r} = r\boldsymbol{e}_r$ to obtain

$$\boldsymbol{v} = \dot r \boldsymbol{e}_r + r\dot\varphi \boldsymbol{e}_\varphi. \tag{32.2}$$

Differentiating both terms once more gives radial terms $\ddot r \boldsymbol{e}_r - r\dot\varphi^2 \boldsymbol{e}_r$ and azimuthal terms $\dot r\dot\varphi \boldsymbol{e}_\varphi + (\dot r\dot\varphi + r\ddot\varphi)\boldsymbol{e}_\varphi$. Thus

$$\boldsymbol{a} = (\ddot r - r\dot\varphi^2)\boldsymbol{e}_r + (r\ddot\varphi + 2\dot r\dot\varphi)\boldsymbol{e}_\varphi. \tag{32.3}$$

Both the scalar components and the basis vectors must be differentiated. Orthonormality also gives $v^2 = \dot r^2 + r^2\dot\varphi^2$.

32.4 Area rate and the running ellipse

For a planar differentiable path $\boldsymbol{r}(t) = (x(t), y(t))$, define its *signed areal velocity* at t as the limit of signed triangle area per time increment, with the origin as the third vertex. Lecture 13's oriented area formula gives that triangle's area as $\frac{1}{2}(x(t)y(t+h) - y(t)x(t+h))$. Subtract the cancelling product $x(t)y(t)$ in the numerator and divide by h. The component derivative limits give

$$\mathcal{A}(t) = \frac{1}{2}\left(x\dot y - y\dot x\right) = \frac{1}{2}r^2\dot\varphi$$

in polar coordinates. The polar equality follows by inserting Eq. (32.2). The radial term cancels, and the angular area factor is $\cos^2\varphi + \sin^2\varphi = 1$.

On a segment where φ is strictly monotone and spans at most 2π, write the radius as a continuous function R of the angle, using the continuous monotone inverse. By Lecture 28, the signed sector area from a fixed time a to t in this segment is

$$A(t) = \frac{1}{2}\int_{\varphi(a)}^{\varphi(t)} R(\eta)^2 \,\mathrm{d}\eta.$$

Reversed endpoints give negative signed area for a decreasing angle. The fundamental theorem and chain rule give $\dot A(t) = R(\varphi(t))^2\dot\varphi(t)/2 = \mathcal{A}(t)$. Thus the triangle-limit definition agrees with the sector formula where that single-sweep geometric description applies.

The azimuthal acceleration coefficient can be written as

$$a_\varphi = r\ddot\varphi + 2\dot r\dot\varphi = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}(r^2\dot\varphi).$$

Since $r > 0$, it vanishes identically on an interval exactly when $r^2\dot{\varphi}$ is constant there, by Lecture 23's constant-function theorem. Equivalently, radial acceleration alone preserves signed areal velocity.

Example 32.2 (Uniform angular speed on the ellipse). Take $r = p/(1 + \varepsilon\cos\varphi)$ with $p > 0$ and $0 \le \varepsilon < 1$, and prescribe $\varphi = \omega t$ with $\omega > 0$. Write $D = 1 + \varepsilon\cos\varphi$, which is at least $1 - \varepsilon > 0$. Differentiating r with respect to the angle gives $r_\varphi = p\varepsilon\sin\varphi/D^2$. Consequently the velocity and speed are

$$\boldsymbol{v} = \frac{\omega p\varepsilon\sin\varphi}{D^2}\boldsymbol{e}_r + \frac{\omega p}{D}\boldsymbol{e}_\varphi, \qquad v = \frac{\omega p\sqrt{1 + 2\varepsilon\cos\varphi + \varepsilon^2}}{D^2}.$$

The square-root numerator follows by expanding $\varepsilon^2\sin^2\varphi + D^2 = 1 + 2\varepsilon\cos\varphi + \varepsilon^2$. The signed areal velocity is $\mathcal{A} = \omega p^2/(2D^2)$. It is constant when $\varepsilon = 0$ and varies with angle when $0 < \varepsilon < 1$. Thus uniform angular speed on this ellipse about the polar origin need not give a constant area rate.

32.5 Tangent, normal and curvature

A C^2 path is *regular* on I if $\boldsymbol{v}(t) \neq \boldsymbol{0}$ for every $t \in I$. For a regular path the speed is positive and C^1, as the square root of the positive C^1 function $\boldsymbol{v}\cdot\boldsymbol{v}$. Its *unit tangent* is $\boldsymbol{T} = \boldsymbol{v}/v$. Differentiating $\boldsymbol{T}\cdot\boldsymbol{T} = 1$ gives $2\boldsymbol{T}\cdot\dot{\boldsymbol{T}} = 0$, so the tangent's derivative is perpendicular to the tangent. Define the nonnegative *curvature* and, where it is positive, the *principal unit normal* by

$$\kappa = \frac{\|\dot{\boldsymbol{T}}\|}{v}, \qquad \boldsymbol{N} = \frac{\dot{\boldsymbol{T}}}{\|\dot{\boldsymbol{T}}\|} \quad (\kappa > 0).$$

The normal is unit and perpendicular to $\boldsymbol{T}$.

Theorem 32.3 (Tangential and normal acceleration). *For a regular C^2 path, wherever $\kappa > 0$,*

$$\boldsymbol{a} = \dot{v}\boldsymbol{T} + \kappa v^2\boldsymbol{N}, \qquad \dot{v} = \frac{\boldsymbol{v}\cdot\boldsymbol{a}}{v}.$$

At a point where $\kappa = 0$, one has $\boldsymbol{a} = \dot{v}\boldsymbol{T}$ and no principal normal is assigned by this definition. In three dimensions, or after embedding a planar path in three dimensions,

$$\kappa = \frac{\|\boldsymbol{v}\times\boldsymbol{a}\|}{v^3}.$$

Proof. Differentiate $v^2 = \boldsymbol{v}\cdot\boldsymbol{v}$ using the dot-product rule. This gives $2v\dot{v} = 2\boldsymbol{v}\cdot\boldsymbol{a}$, and division by $v > 0$ proves the speed formula. Next differentiate $\boldsymbol{v} = v\boldsymbol{T}$: $\boldsymbol{a} = \dot{v}\boldsymbol{T} + v\dot{\boldsymbol{T}}$. Where $\kappa > 0$, the definition gives $\dot{\boldsymbol{T}} = \kappa v\boldsymbol{N}$, proving the decomposition. If $\kappa = 0$, then $\dot{\boldsymbol{T}} = \boldsymbol{0}$, so only the tangential term remains.

For positive curvature, cross the decomposition with $\boldsymbol{v} = v\boldsymbol{T}$. The tangent term has zero cross product, giving $\boldsymbol{v}\times\boldsymbol{a} = \kappa v^3(\boldsymbol{T}\times\boldsymbol{N})$. The two unit vectors are perpendicular, so their cross product has norm 1 by Lecture 13. Taking norms proves the curvature formula. At zero curvature the acceleration is parallel to velocity, so the cross product is zero and the same formula holds. ∎

For a planar path, the embedded cross product has only third component $\dot{x}\ddot{y} - \dot{y}\ddot{x}$. Hence $\kappa = |\dot{x}\ddot{y} - \dot{y}\ddot{x}|/(\dot{x}^2 + \dot{y}^2)^{3/2}$, with the regularity hypothesis retained.

For a regular motion on an interval, the speed formula and Lecture 23 show that speed is constant exactly when $\boldsymbol{v} \cdot \boldsymbol{a} = 0$ at every time. A nonzero acceleration can therefore coexist with constant speed, provided it is perpendicular to velocity. Zero acceleration instead makes the velocity constant componentwise. The reconstruction formula then gives $\boldsymbol{r}(t) = \boldsymbol{r}_0 + (t - t_0)\boldsymbol{v}_0$. For nonzero $\boldsymbol{v}_0$, this straight motion has constant tangent and zero curvature.

Curvature is unchanged by a regular change of parameter. Let J be an open interval and $\psi : J \to I$ a C^2 map with $\psi' > 0$. The reparametrised path is $\widetilde{\boldsymbol{r}}(\tau) = \boldsymbol{r}(\psi(\tau))$. Its velocity is $\boldsymbol{v}(\psi)\psi'$, its speed is $v(\psi)\psi'$, and its unit tangent is $\boldsymbol{T}(\psi)$. The derivative of that tangent has norm $\left\|\dot{\boldsymbol{T}}(\psi)\right\| \psi'$. The quotient defining curvature cancels the positive factor, giving $\widetilde{\kappa}(\tau) = \kappa(\psi(\tau))$. If instead $\psi' < 0$, the new tangent reverses sign; its derivative norm and the new speed both acquire $|\psi'|$, so the same equality holds. These calculations use the componentwise chain rule, with the indicated strict derivative sign on the parameter interval.

The signed arc-length coordinate from a fixed t_0 is $s(t) = \int_{t_0}^{t} v(u)\,\mathrm{d}u$. The fundamental theorem gives $\dot{s} = v > 0$, so it is strictly increasing. Locally the inverse derivative rule permits s as a parameter with $\mathrm{d}t/\mathrm{d}s = 1/v$. Consequently $\kappa = \|\mathrm{d}\boldsymbol{T}/\mathrm{d}s\|$: curvature measures turning per unit distance. The one-chord partition in Lecture 28 also gives $\|\boldsymbol{r}(b) - \boldsymbol{r}(a)\| \le \int_a^b v(t)\,\mathrm{d}t$ for $a < b$ in I.

For the general projectile, the planar cross-product component is $-gu$. Since $g, u > 0$, its curvature is

$$\kappa(t) = \frac{gu}{\left(u^2 + (w - gt)^2\right)^{3/2}}.$$

The speed squared is $u^2 + (w - gt)^2$, so it is smallest at the highest point $t = w/g$. Curvature is therefore largest there, with value g/u^2. At that instant the speed is u, the tangential acceleration is zero, and the normal acceleration coefficient is $\kappa v^2 = g$. The acceleration magnitude remains g throughout the flight even though its tangent and normal components change. Before the highest point $\dot{v} = -g(w - gt)/v < 0$; afterwards it is positive. This separates a change in speed from a change in the velocity direction. Curvature has units of inverse length, making κv^2 an acceleration.

Figure 32.3 resolves the running projectile's acceleration at one rising point. Its components are added head to tail.

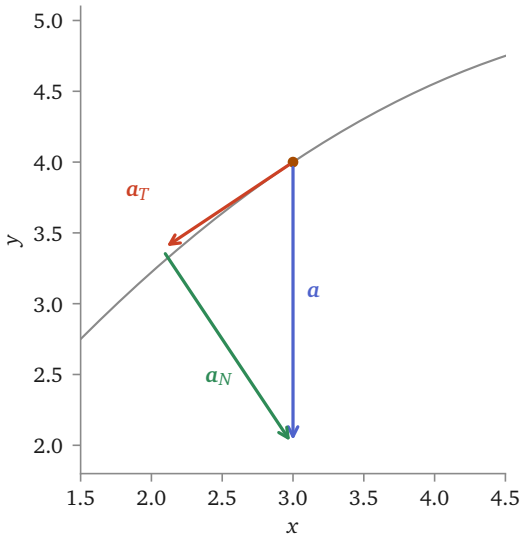


Figure 32.3: At $t = 1$ for the numerical projectile, tangential and normal acceleration components sum to the downward acceleration. All three acceleration arrows use one common display scale.

Example 32.4 (Resolving the projectile acceleration)**.** For $u = 3$, $w = 4$, $h = 1$ and $g = 2$, resolve the acceleration at $t = 1$. The velocity is $(3, 2)$, acceleration is $(0, -2)$ and speed is $\sqrt{13}$. The formulas give

$$\dot{v} = -\frac{4}{\sqrt{13}}, \qquad \kappa = \frac{6}{13\sqrt{13}}, \qquad \boldsymbol{T} = \frac{(3,2)}{\sqrt{13}}, \qquad \boldsymbol{N} = \frac{(2,-3)}{\sqrt{13}}.$$

For the normal, differentiate $\boldsymbol{v}/v$ or normalise $\boldsymbol{a} - \dot{v}\boldsymbol{T} = (12/13, -18/13)$. The two acceleration vectors are therefore

$$\boldsymbol{a}_T = \dot{v}\boldsymbol{T} = (-12/13, -8/13), \qquad \boldsymbol{a}_N = \kappa v^2 \boldsymbol{N} = (12/13, -18/13).$$

Their sum is $(0, -2)$, recovering the prescribed acceleration. The negative speed derivative expresses slowing during ascent; the positive normal coefficient expresses the change in direction.

The hypotheses distinguish two exceptional cases. If the projectile model is changed to $u = 0$, its velocity vanishes at $t = w/g$. The normalised tangent is then undefined, even though Cartesian position and acceleration remain well defined. For a separate regular path $\boldsymbol{r}(t) = (t^3 + t, 0)$, the velocity is $(3t^2 + 1, 0)$ and the unit tangent is constantly $(1, 0)$. Thus $\kappa = 0$ and $\boldsymbol{a} = (6t, 0) = \dot{v}\boldsymbol{T}$; a principal normal is not defined, while tangential acceleration can be nonzero. This path also crosses the polar origin at zero. Its nonnegative radius $|t^3 + t|$ has left derivative -1 and right derivative 1 at zero. Thus the smooth polar representation hypotheses fail even though the Cartesian path is regular.

Summary 32.

- Component limits define vector derivatives. Continuous prescribed acceleration and initial position/velocity determine one C^2 motion by Eq. (32.1).
- Equations (32.2) and (32.3): a positive C^2 radius and a continuous C^2 angle give the polar velocity and acceleration, including both basis derivatives.
- Signed areal velocity is $(x\dot{y} - y\dot{x})/2 = r^2\dot{\varphi}/2$ in polar coordinates. Vanishing azimuthal acceleration makes it constant on an interval with $r > 0$.
- Theorem 32.3: for a regular C^2 path, curvature is $\|\boldsymbol{v} \times \boldsymbol{a}\| / v^3$. Where $\kappa > 0$, acceleration is $\dot{v}\boldsymbol{T} + \kappa v^2 \boldsymbol{N}$; where $\kappa = 0$, it is $\dot{v}\boldsymbol{T}$.
- Zero speed prevents the normalised tangent construction. A polar-origin crossing needs its own coordinate check; regular Cartesian motion can continue there.

Sheet 32 treats nonconstant acceleration, space curves, polar motion and the ellipse with constant areal velocity. Lecture 33 asks how to solve first-order equations when the unknown function also occurs in its derivative rule.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§13.1–13.4.